

AUSTO MOTOR COMPANY ANALYSIS

PROJECT REPORT

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1. Data Description and Data Dictionary:

The data contains the details for the austo motors car manufacturer specializing in SUV, Sedan, and Hatchback models. We are going to check how all these variables are influencing the Make of company.

Variable Name	Data Types	Description
Age	Int 64	The age of the individual in years.
Gender	object	The gender of the individual, categorized as male or female.
Profession	object	The occupation or profession of the individual.
Marital_status	object	The marital status of the individual, such as married &, single.
Education	object	The educational qualification of the individual Graduate and Post Graduate.
No_of_Dependents	Int 64	The number of dependents (e.g., children, elderly parents) that the individual supports financially.
Personal_loan	object	A binary variable indicating whether the individual has taken a personal loan "Yes" or "No"
House_loan	object	A binary variable indicating whether the individual has taken a housing loan "Yes" or "No"
Partner_working	object	A binary variable indicating whether the individual's partner is employed "Yes" or "No"
Salary	Int64	The individual's salary or income.
Partner_salary	Float64	The salary or income of the individual's partner, if applicable.
Total_salary	Int64	The total combined salary of the individual and their partner (if applicable).
Price	Int64	The price of a product or service.
Make	object	The type of automobile

Table 1 Data Description and Data Dictionary

- There are 14 Variables (columns) and 1581 rows in this dataset
- Out of 14 variables 5 of them are of int64 type, 1 of them are of float data types and 8 variables are of object data types
- There is a missing value present in the Gender and Partner_salary column, which have been imputed with making separate category i.e. Unknown_gender and I have imputed the partner salary with mode of Partner_salary = (Total_salary – Salary).

2. Data Overview:

List of Libraries

- Used pandas and NumPy libraries, making alias pd and np respectively so that we can read and manipulate the dataset i.e austo motors company.
- For Visualization, we have imported matplotlib (from matplotlib we have imported pyplot as plt) and seaborn libraries as sns.
- Now we have restricted the float value up to 2 decimal places in the dataset.

2.1 Importing the dataset:

- Used code to read the csv file and store the dataset into df

2.2 View First and last five rows:

We have used code df. head () to see the first five rows of the dataset df and for the last five rows of dataset we can use df. tail ()

Observations:

There are 1581 rows and 14 columns are present in the dataset df

2.3 Information about the dataset df

Checking the datatypes of the columns, non-null value count, memory allocation for the csv file and data types of each column.

Observations:

- There are 6 numerical columns (1 column is of float data types and 5 columns are of int data type) in the data set and 8 are of object data types of columns in dataset df.
- There are columns which are having less entries than the other columns which indicates the presence of missing values in the dataset df specially in column (Gender and Partner_salary column)
- The memory uses are 173.0 + KB

2.4 Duplicate value in dataset df by using df. duplicated (). sum ()

Observation:

There are zero duplicates value in dataset df

2.5 Unique Value Check for all categorical Variable:

```
☞ The Unique value of Gender are : ['Male' 'Femal' 'Female' nan 'Femle']
.....
The Unique value of Profession are : ['Business' 'Salaried']
.....
The Unique value of Marital_status are : ['Married' 'Single']
.....
The Unique value of Education are : ['Post Graduate' 'Graduate']
.....
The Unique value of Personal_loan are : ['No' 'Yes']
.....
The Unique value of House_loan are : ['No' 'Yes']
.....
The Unique value of Partner_working are : ['Yes' 'No']
.....
The Unique value of Make are : ['SUV' 'Sedan' 'Hatchback']
.....
```

Figure 1 Unique value for categorical variable

Observation:

- Female word has been written in 3 different spelling, “Femal”, “Femle” and Female.
- We need to replace the “Femal”, “Femle” with correct word i.e. Female, so that the distribution of gender will show the correct result.
- Rest all the categorical variable is having the correct unique value.

Correcting the spelled word “Femal”, “Femle” with “Female.”

```
# as female word written in 3 different spelling, replacing the Femal and Femle data with correct spelling Female
df["Gender"] = df["Gender"].replace({"Femal": "Female", "Femle": "Female"})
print(df["Gender"].unique())

['Male' 'Female' nan]
```

Figure 2 Correction of misspelled word of Female in Gender column

- With the help of replace method we have replaced the wrong spelling. And we can see that there are three categories of Gender (Male, Female, nan).
- There are some missing values in gender column which is in the form of Nan.

2.6 Checking null values in dataset df:

	Null Value Count	% of Null Value
Gender	53	3.35
Partner_salary	106	6.70

Figure 3 Checking null values in data set

Observation:

- After calculation of number of null value and their percentage, there are 53 missing values in gender column i.e. (3.35%) and 106 value are missing in partner_salary column i.e. (6.70%) out of 1581 data.
- The missing values will be treated after understanding the distributions of features in the data, if any relationship that exist in the data. This will help us impute these values more effectively.

2.7 Treatment of missing value in Gender column:

```
Gender
Male    0.76
Female  0.21
NaN     0.03
Name: proportion, dtype: float64
```

Figure 4 % of Missing value in Gender column

Observation:

- As we can see that there is missing value 3% in Gender Column

- The percentage of missing values is very less than the percentage of observations from another Gender.
- As the percentage of missing value in gender out of 1581 is only 3.35%. It is less than the 5%. So, we can treat this missing value with the help of three methods.
 - I. With the help of mode.
 - II. We can remove the Nan value from the dataset
 - III. We can make separate category of Nan Value

```
[13] # Replacing NaN with Unknown
df['Gender'].fillna('Gender_Unknown', inplace = True)

[14] # Treating missing value in Gender column
df["Gender"].value_counts(normalize = True, dropna = False)

Gender
Male      0.76
Female    0.21
Gender_Unknown 0.03
Name: proportion, dtype: float64

[15] df["Gender"].unique()

array(['Male', 'Female', 'Gender_Unknown'], dtype=object)
```

Figure 5 Filling the Nan value with gender unknown in Gender Column

- We can treat the missing values with a separate category for this variable as it is very less and it won't affect the dataset.
- We can fill "Gender-unknown" value in place of Nan value.
- After replacement of Nan value with Gender_Unknown value, new label has been updated in Gender column.

2.8 Treatment of missing value in Partner_salary column:

- There are 106 observations (6.70 %) where 'Partner_salary' variable has missing values from the overview of the dataset.
- Since Partner_salary is a continuous numerical variable, so we can impute the missing values by mean or median but before imputation, let's analyse the Partner_salary variable in detail.

As we can see that the partner_Salary variable is related to Total salary

$$\text{Total_Salary} = \text{Salary} + \text{partner_Salary (column)}$$

- In all the Salary column there is values, but in partner_salary there are sum missing value also. So, Partner_salary = Total_salary- salary we can do in case of if partner is working. In case of Partner not working then automatically it will assign 0.
- We can check in which row the missing value is present in the partner salary, so that we can find the relationship with any other columns, if exist.

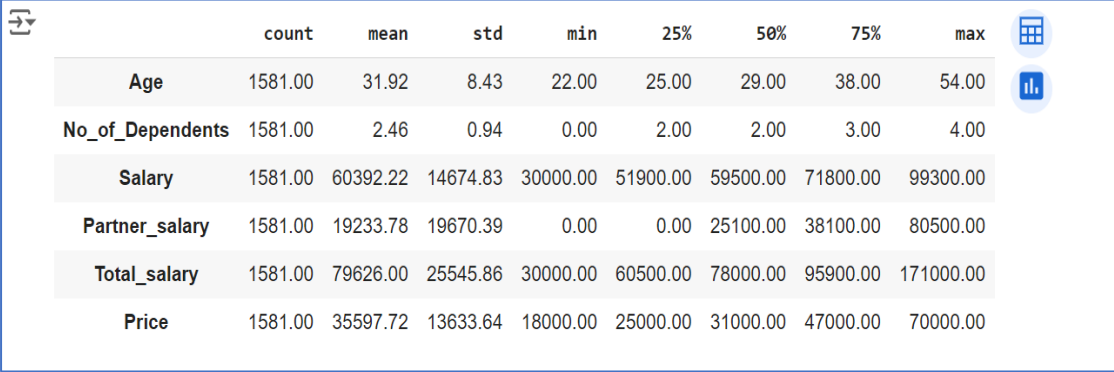
```
[18] # Calculate Partner_salary for missing values using Total_salary - Salary
df.loc[df['Partner_salary'].isnull()==True, 'Partner_salary'] = df['Total_salary'] - df['Salary']
```

Figure 6 printing the rows which is having Nan Value in Partner Salary

- We will calculate the partner_salary for missing values by using Total_salary – Salary. It helps in maintaining the data completeness.
- Now the dataset is fully cleaned.

3. Statistical summary of the data

The statistical summary provides the insight of distribution of column for each feature in the dataset df.



	count	mean	std	min	25%	50%	75%	max
Age	1581.00	31.92	8.43	22.00	25.00	29.00	38.00	54.00
No_of_Dependents	1581.00	2.46	0.94	0.00	2.00	2.00	3.00	4.00
Salary	1581.00	60392.22	14674.83	30000.00	51900.00	59500.00	71800.00	99300.00
Partner_salary	1581.00	19233.78	19670.39	0.00	0.00	25100.00	38100.00	80500.00
Total_salary	1581.00	79626.00	25545.86	30000.00	60500.00	78000.00	95900.00	171000.00
Price	1581.00	35597.72	13633.64	18000.00	25000.00	31000.00	47000.00	70000.00

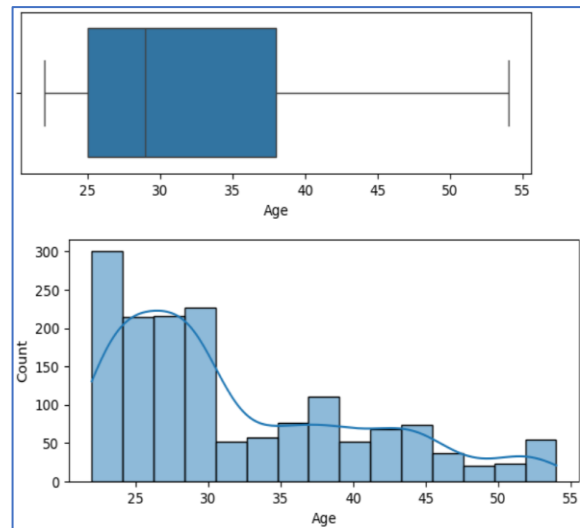
Figure 7 Statistical Summary of Numerical Data

- **Age:** On an average age of an individual is 31 yrs., 50% of the individual are of 29 yrs. in this dataset. 75% of an individual having age less than or equal to 38 yrs. The minimum and the maximum age of an individual are 22 yrs. and 54 yrs. respectively.
- **No_of_Dependents:** There are 0 no of dependents and few of the dependents is having up to 4 dependents. On an average there are 2 no of dependents of customer are available in dataset. 75% of an individual are having less than or equal to 3 dependents.
- **Salary:** The range of salary from 30,000 to 99,300. 50 % of salary distribution are 59,000. 75% of the salary are less than or equal to 71,800. It is noticeable that there are diverse income levels of the customer.
- **Partner_salary:** The minimum partner salary is showing 0 which indicates many partners may not earn salary. Mean of partner_salary is very less; it shows many partners either does not work or they earn very less salary. On an average partner is earning 25,100. 75% of partner salary is less than or equal to 38,100. The maximum partner salary is 80,500.
- **Total_salary:** It reflects the total household earning i.e. salary + partner_salary. On an average the total salary is 79,262. The range of total salary are 30,000 to 171,000. 50% of the total salary lies at 78,000. There is a wide range of salary due to salary of an individual and partner salary.
- **Price:** on an average the price of car is 35,597. Median of price of car is 31,000. The range of car price is in between 18,000 to 70,000. 75% of car is having price 47,000.

4. Exploratory Data Analysis: Univariate Analysis (Visualization Method)

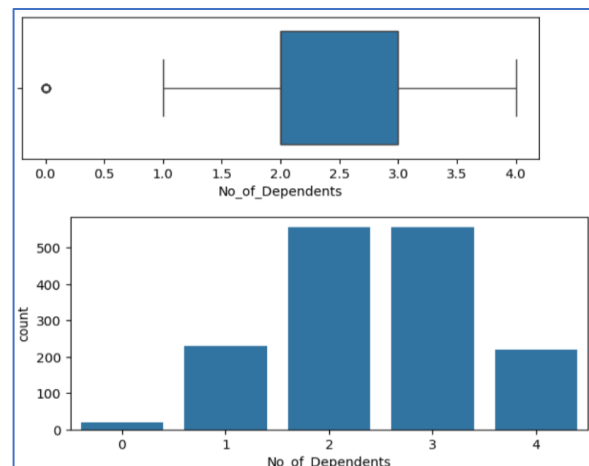
4.1 Observations: Age

- The distribution of Age is little right skewed.
- There are no outliers present in the Age column, that means, the age is range.
- The age group where customer buys car is in between 20 yrs. to 54 yrs. 50% of the data of Age lies under 30 yrs. It indicates more customer buying the car who has age less than 30 yrs.
- The age of customer are in between 25 to 38 yrs. And there are few customers who is having age 54 yrs.



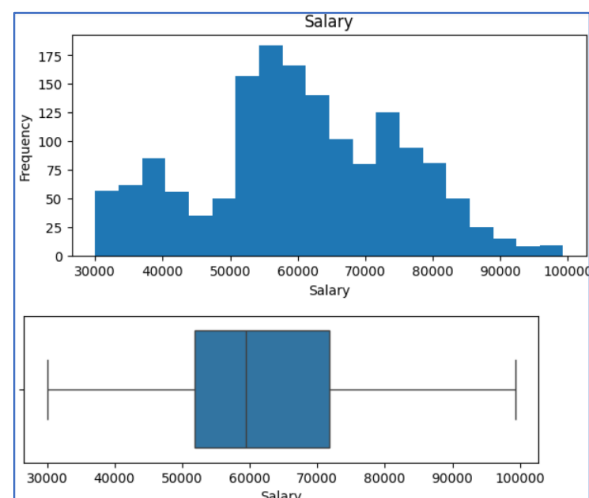
4.2 Observations: No of Dependents

Approximately 230 customers are there who are having no of dependents 1. A significant number of customers who are having no of dependents 2 are more than 500. There are approximately 25 customers who are not having any dependents. There are also approximately 250 customers who are having no of dependents 4.



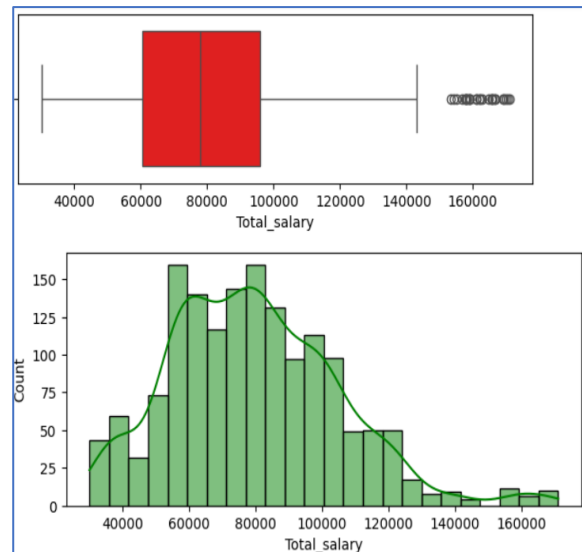
4.3 Observations: Salary

- The distribution of salary is slightly uniform.
- There are no outliers present in the salary column.
- 50% of buyers who are having salary are less than or equal to 60,000, which indicates, customer spending more money on buying car.
- Most of the buyers are having salary in the range of 55,000 to 75,000.
- There are very few customers who are having salary greater than 90,000.



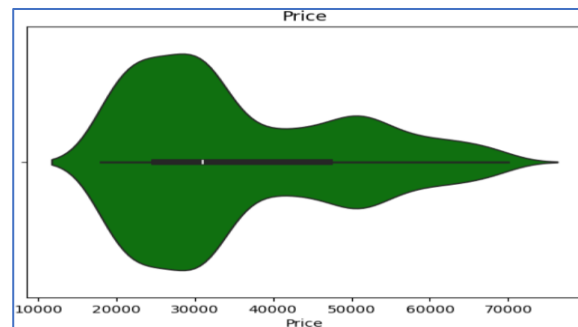
4.4 Observations: Total_salary

- There are outliers present in the Total_Salary column.
- The distribution of the Total_salary of customer is little right skewed. It indicates that 50% of customers are having salary less than or equal to 80,000.
- Most of the buyers are having Total_salary in the range of 60,000 to 1,00,000.
- There are very less customers who is having salary greater than 1,40,000.



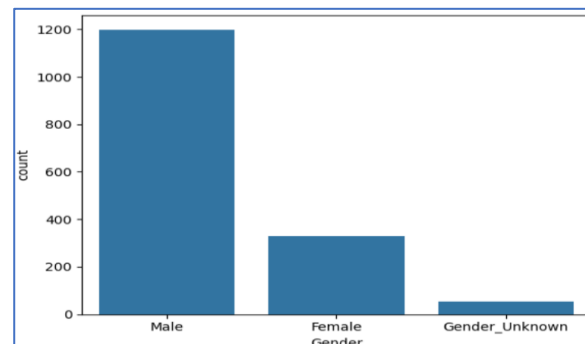
4.5 Observations: Price

- On an average the price of car is 35597.
- Median of price of car is 31,000.
- The range of car price is in between 18,000 to 70,000.
- Most of the car price lies between 31,000 to 47,000



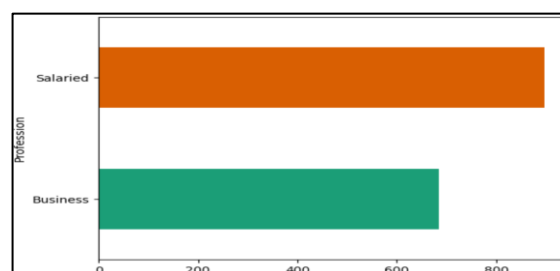
4.6 Observations: Gender

- There are approximately 1150 males and 280 females are there in the Gender column in dataset
- The number of males is more than the female in gender category. We can say that male is buying more car as compare to females.



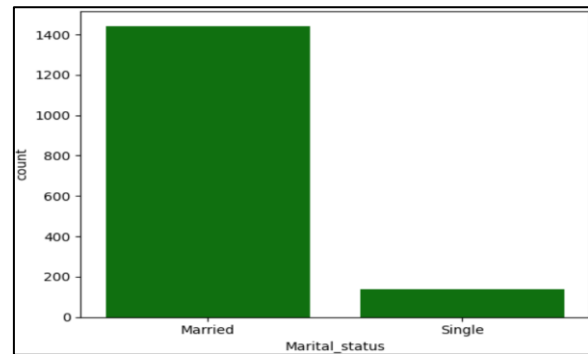
4.7 Observations: Profession

- As we can see that business professionals are less as compared to salaried professionals.
- Salaried professionals buy more car than the business professionals.



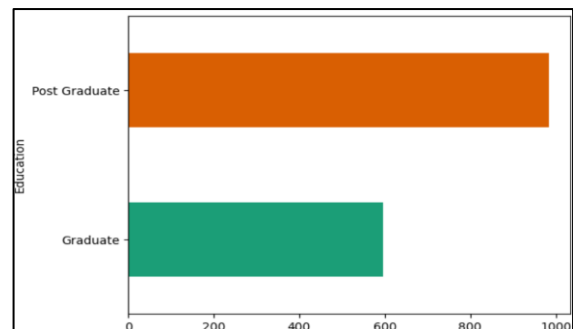
4.8 Observations: Marital_status

- There are more than 1400 married who has bought the cars, whereas approx. 180 singles have bought the car.
- As compare to singles, married people are buying more car.



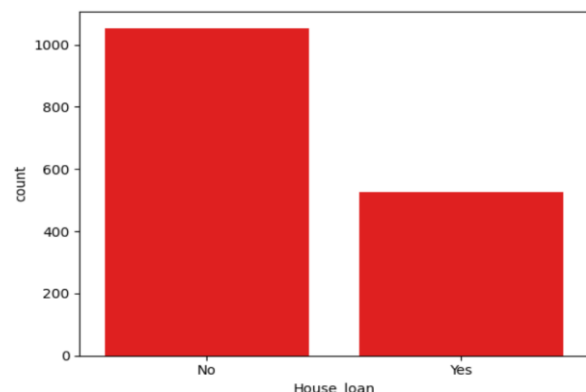
4.9 Observations: Education

- Number of postgraduates is more as compare to the Graduate in the dataset df.
- We can say that there is customer who is buying the car are of education postgraduation



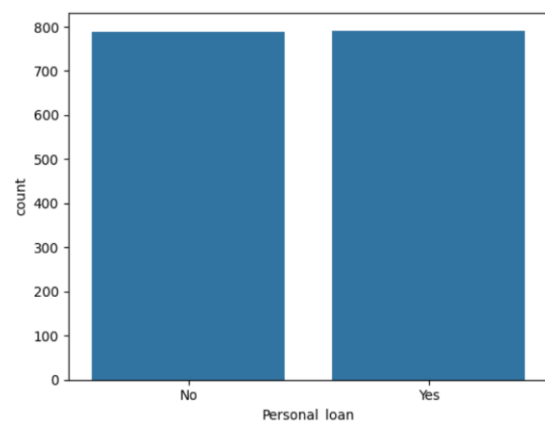
4.10 Observations: House_loan

- The total number of customers who has taken house loan is approx. 500 and number of customers who has not taken house loan are 1100.
- The customer who has not taken house loan is almost two times of customer who has taken house loan.



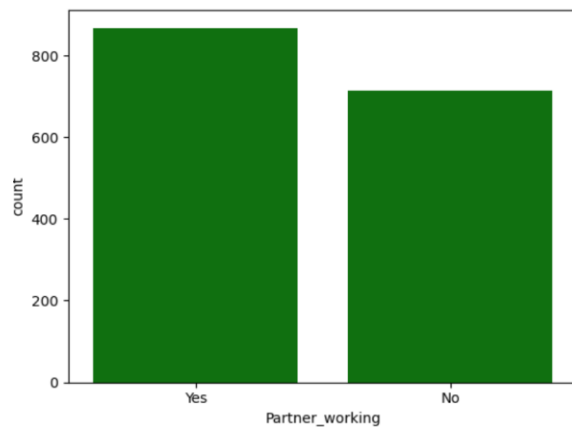
4.11 Observations: Personal_loan

- There is 1:1 ratio of customer who has taken personal loans or who has not taken the personal loan. (50% - 50%)
- From the data set visuals, we can say that the number of customers who is having personal loan is equal to the customer who is not having any personal loans.



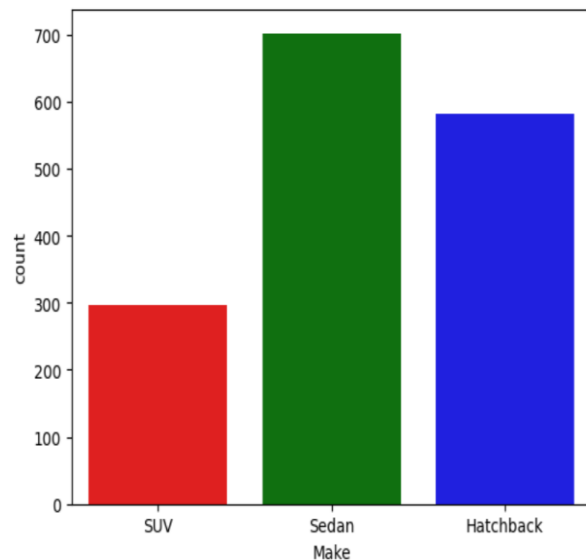
4.12 Observations: Partner_Working

- The number of working partners is more as compare to non-working partner.
- Working partner is more then the total salary of a customer is more.
- It indicates that if customer has more Total salary, then very high chance to spend more money on buying the car



4.13 Observations: Make

- The maximum cars sold make type Sedan.
- Lowest number of cars sold make type SUV, it indicates that customer is preferring Sedan, hatchback as compare to SUV. As price of SUV is more as compare to other Make type car.
- More than 700 customer is buying sedan, 550 customers are buying hatchback where as the number of customers who is buying SUV are approx. 300.



5. OUTLIER DETECTION:

For outlier detection we have created a separate category for numerical column of data and add their name in the list. And let's Visualise it through box plot

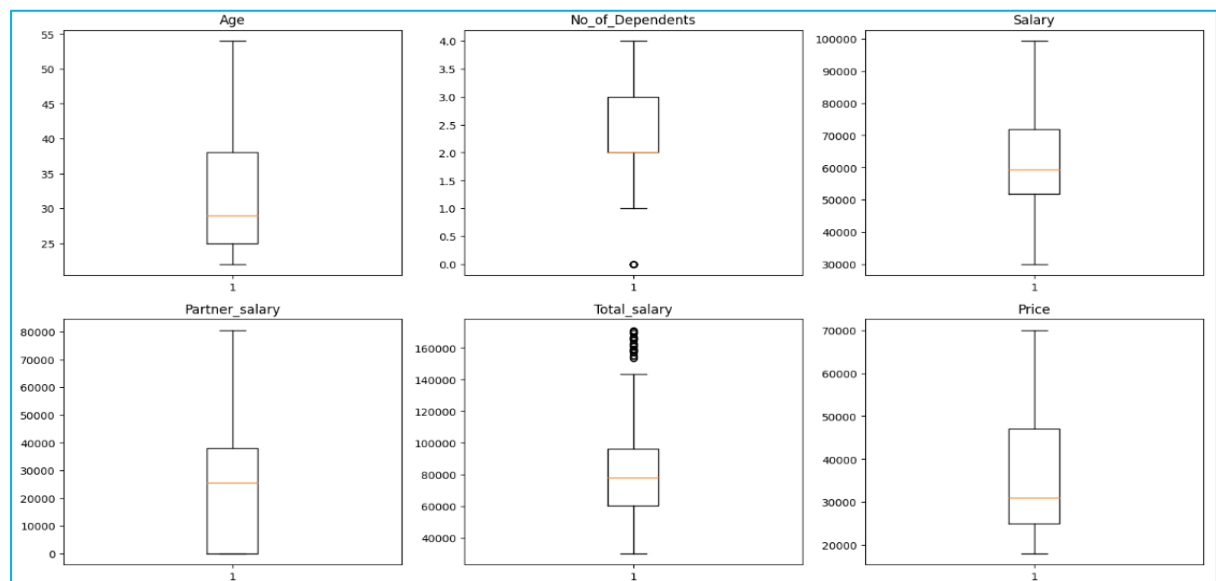


Figure 8 Outlier Detection

Observation:

We have used box plot which provides the visual representation of the data distribution. We have got the outlier in two numerical columns. 1. No of dependent, 2. Total_Salary.

Now let's calculate the outliers present in the variable data column

- We will find the 25th percentile and 75th percentile of the numerical column
- Now find the Inter Quantile Range (75th percentile – 25th percentile)
- File the Lower and upper whiskers for all the value.
- The value which we will find outside the whiskers we will consider as outliers

```
Age          0
No_of_Dependents  20
Salary       0
Partner_salary  0
Total_salary  27
Price        0
dtype: int64
```

Figure 9 Showing the number of outlier present in the data

Observations:

As visually and by using calculation we can say that there is an outlier present in the No_of_dependent and Total_salary columns. Total outlier is present in No_of_dependent is 20 and 27 are present in Total_salary columns.

Treatment of outlier:

- All the values smaller than lower_whisker will be assigned the value of lower_whisker
- All the values greater than upper_whisker will be assigned the value of upper_whisker
- Use clip method to remove the outliers.

After treating the outlier, checking the variable by box plot visualization

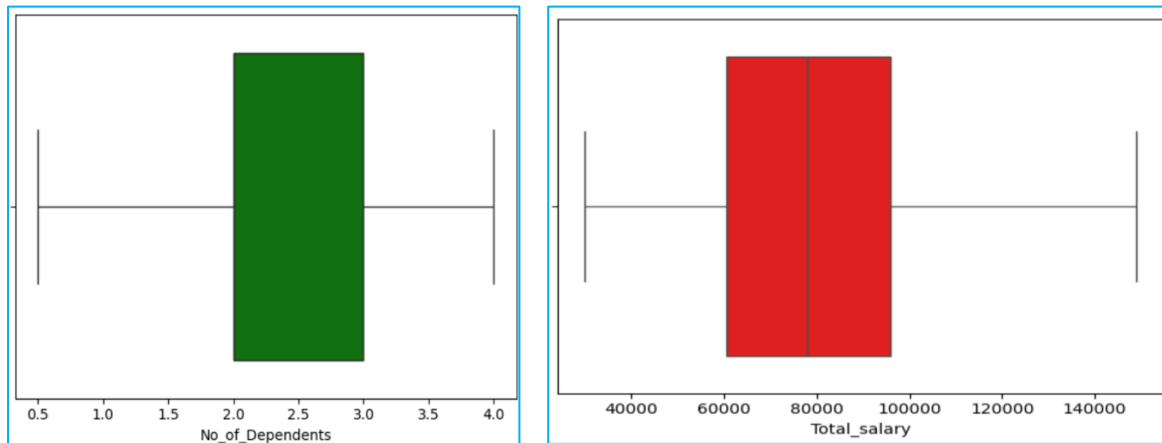


Figure 10 No_of_dependent and totatl_salary column without outlier

6. Exploratory Data Analysis: Bivariate Analysis

Heatmap:

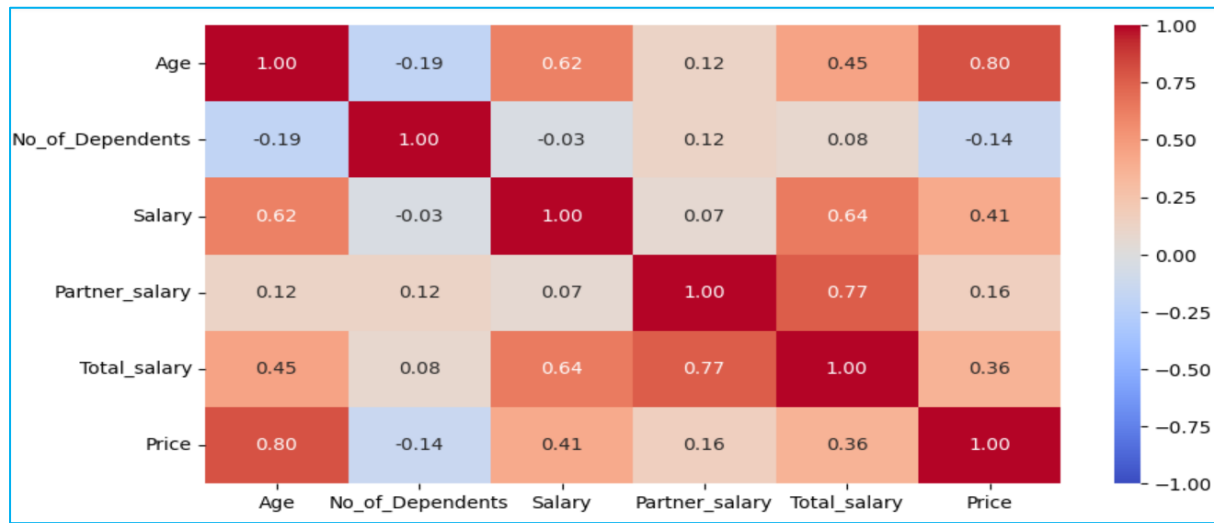


Figure 11 Correlation between all the numerical variables in the dataset

6.1 Observation:

- We have given the correlation parameters from -1 to 1.
- Each column is showing very high correlation with respect to self.
- Price column shows a very high correlation with Age, Salary and Total_salary. It indicates that more the Age, Salary and total salary, customer can spend more money on buying the car.
- Price has a negative correlation with No_of_dependents column. This indicates No_of_dependents increase the customer can spend less money on buying car
- Total_Salary column shows a very high correlation with Age, Salary and Partner_salary. It indicates that more the Age, Salary and partner_salary, The Total_salary will be more
- Salary and Age are having very high correlation. It indicates if customer is older his salary will increase.
- No_of_dependents showing negative correlation with Age. It indicates more the age, no od dependents will be less.

6.2 Make Vs Price Observation:

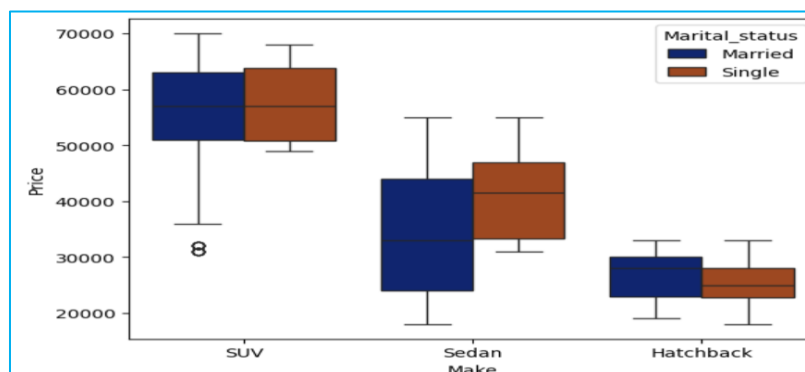


Figure 12 Make Vs Price Observation

- Male customer has purchased SUV as compared to female customers which is having higher price.
- The male customer who is single has bought higher price Sedan car as compared to married male customer.
- In case of Female customer, single has bought lower price Sedan car as compared to married female customer.
- Very less single customer has purchased the hatchback car as compared to married customer.

6.3 Gender buying the car: Observation

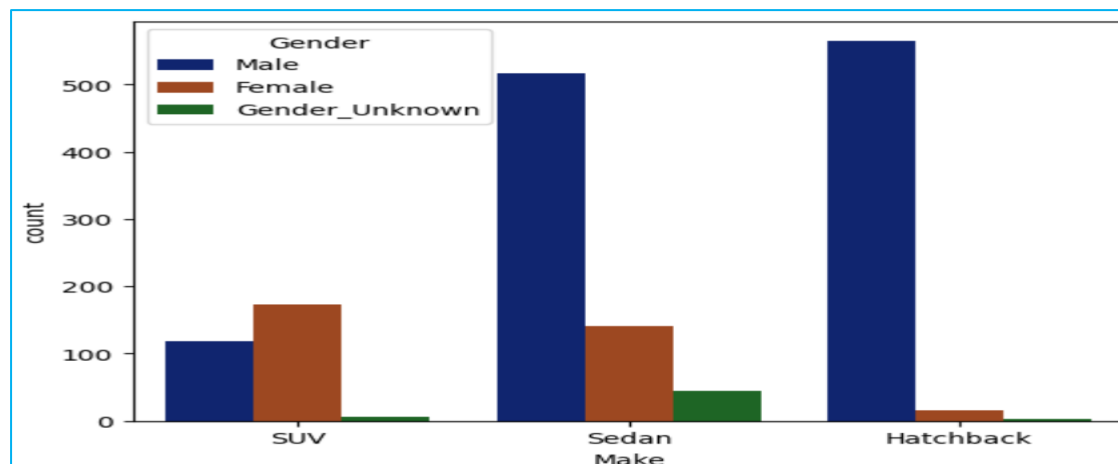


Figure 13 Types of car Gender is buying

- The number of Female customers is buying more SUV car as compare to Male customer.
- Male customer buying more sedan car whereas female customer is buying very less sedan car
- The number of male customers who is buying Hatchback car is more than 500, whereas approx. 75 females are only buying the hatchback car.

6.4 Professionals buying the car: Observation

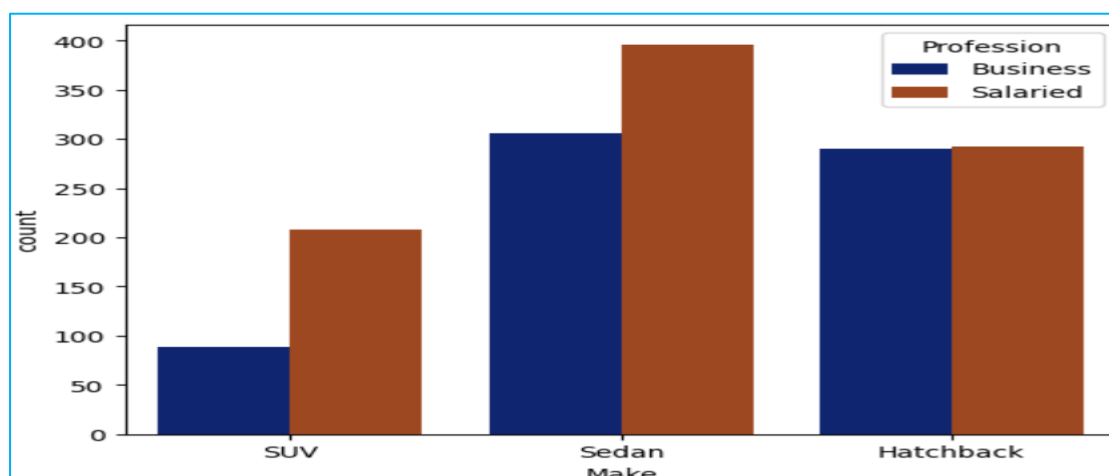


Figure 14 Professionals buying the car

- Salaried customer has bought more SUV as compare to the Business professionals.
- In case of sedan, Number of salaried people who is buying is 400 which is more as compare to the business professionals who is purchasing sedan 300.
- Business professionals and salaried professionals both are buying hatchback equally.

6.5 Pair Plot: Between all the numerical column:

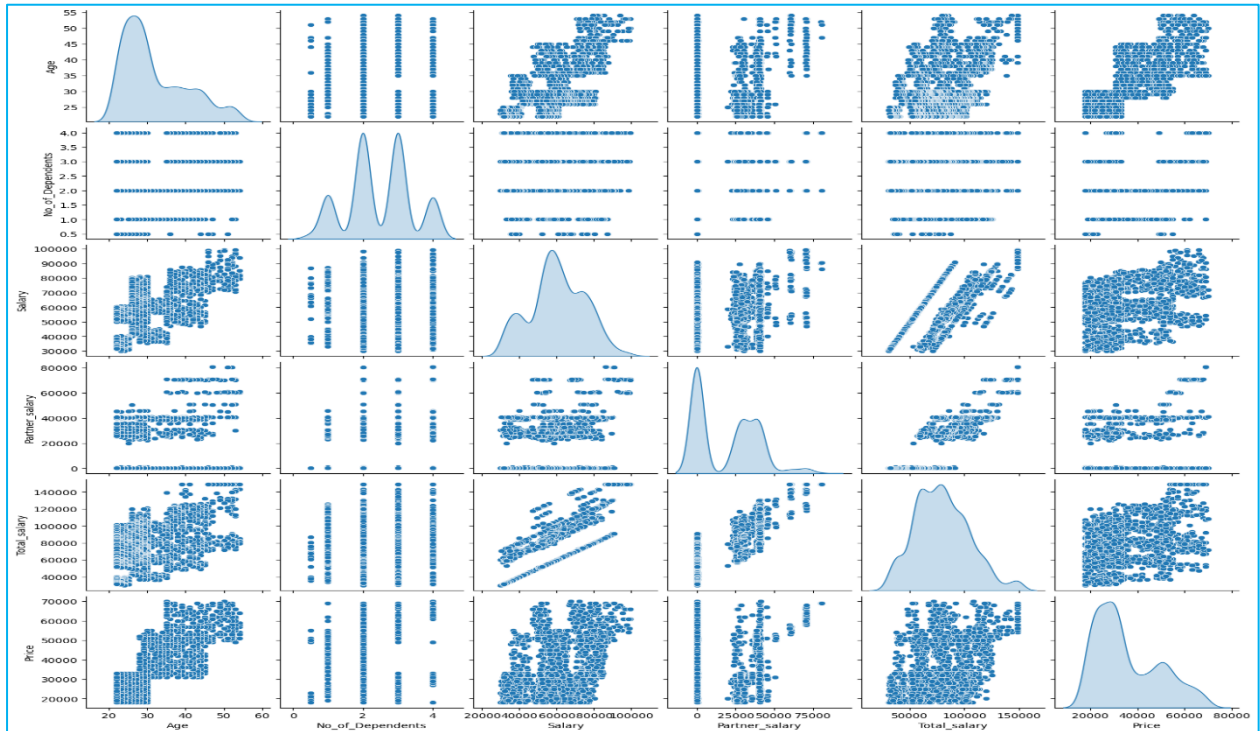


Figure 15 Pair plot between all the numerical column

Observation:

- The diagonal plots showing the numerical variables, and all are in histogram
- Scatter plot showing the relationship between numerical variables in pair.
- “Salary” and “Total_Salary” Is showing positive linear relationship in pair. It is because Total_Salary includes salary + partner_Salary.
- Age vs Total_salary pair scatter plot showing the positive correlation.
- Price of car vs other numerical variables scatter plot shows if there are any trends or patterns between the price of the automobile and other financial variables.

6.6 Total_Salary vs Salary

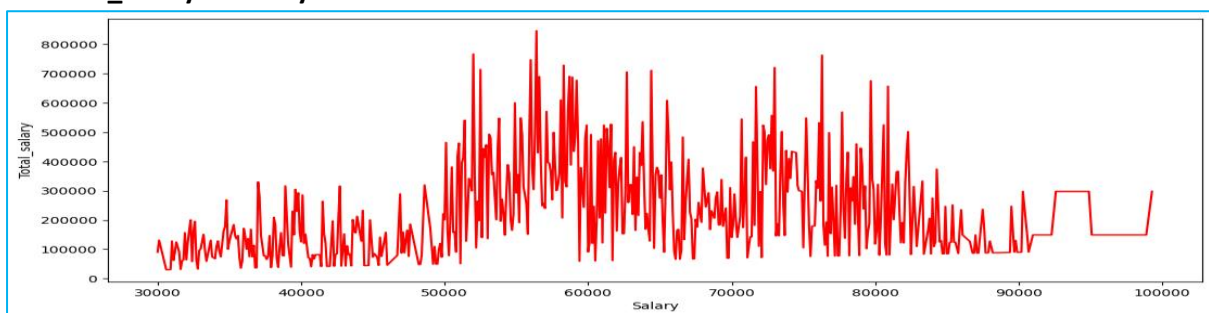


Figure 16 Total_Salary vs Salary

Observation:

- “Salary” and “Total_Salary” Is showing positive linear relationship in pair. It is because Total_Salary includes salary + partner_Salary.
- The total salary is high where the salary of customer is high

7. Multivariate Analysis

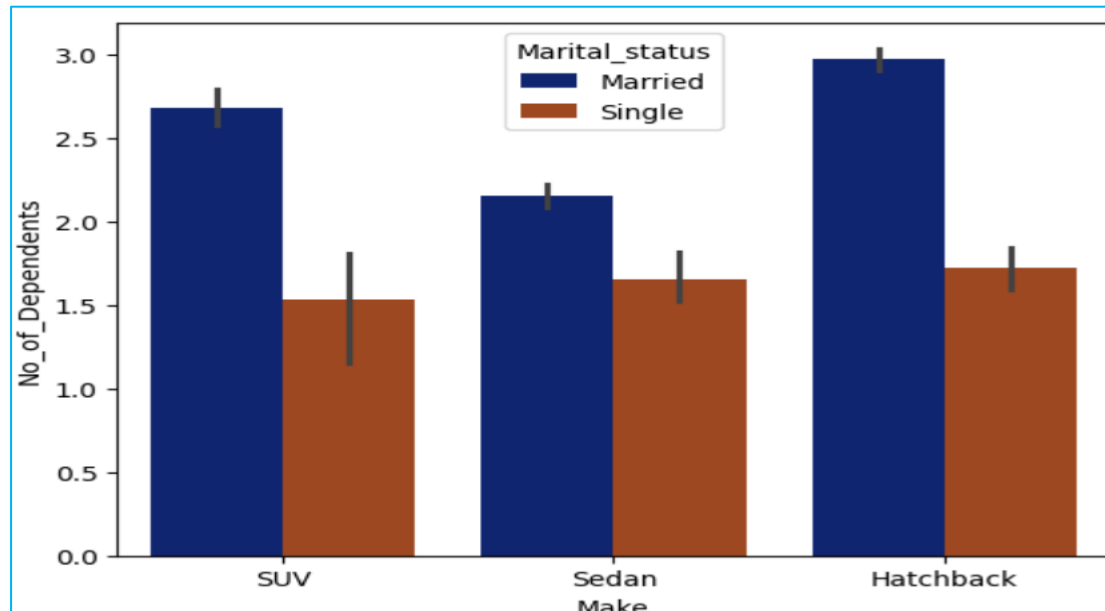


Figure 17 Multivariate analysis

- Customers who are married and having more than 2 no of dependent are mostly buying the hatchback car.
- No customer who is single and number of dependents is more than 2 are buying any car.
- The single customer who is having 1 no of dependent are only buying SUV and sedan.
- Married customer is buying the car more as compare to single. It indicates that their partner is working.

8. KEY Question

8.1 Do men tend to prefer SUVs more compared to women?

	Gender	count	total_count	SUV_Prefer_percentage
0	Female	173	329	52.58
1	Gender_Unknown	6	53	11.32
2	Male	118	1199	9.84

Figure 18 Analysis on preference

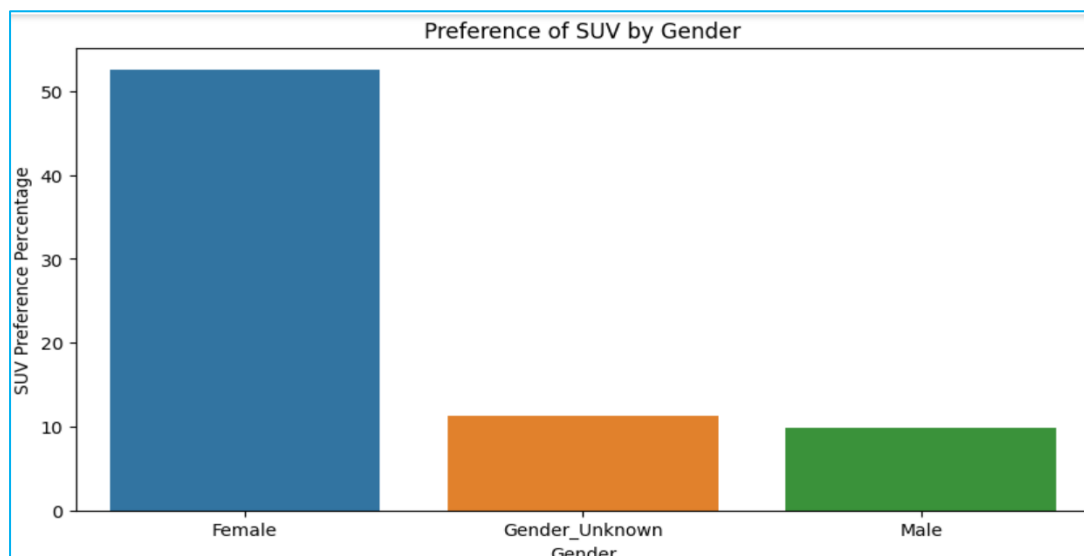


Figure 19 Preference of SUV by Gender

Answer:

After analysis through the graph and calculating the percentage. There is 52.58% Female customer is preferring SUV car whereas the percentage of male customer are only 9.84%. so, we can conclude that no men tend to prefer SUVs more compared to women. In Graph also we can see that female customer prefers SUV car as compare to male.

8.2 What is the likelihood of a salaried person buying a Sedan?

Answer: The likelihood of a salaried person buying a Sedan is approximately 44.20%

8.3 What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

Answer: Observation:

- * Salaried male preferring SUV: 85
- * Percentage of Salaried male preferring SUV: 13.28125
- * Salaried male preferring Sedan: 279
- * Percentage of Salaried male preferring Sedan: 43.59375

Salaried male preferring SUV : 85
 Percentage of Salaried male preferring SUV : 13.28125
 Salaried male preferring Sedan : 279
 Percentage of Salaried male preferring Sedan : 43.59375

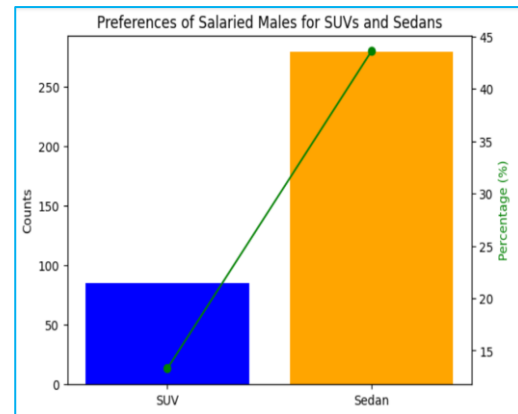


Figure 20 Preference of salaried males for SUV over Sedan

The statement is false as the salaried man is easier target for sedan car as compare to SUV car. The evidence I have posted visually and data supports that Sheldon Cooper's claim is false. The observation found by Sheldon Cooper is completely not valid.

8.4 How does the amount spent on purchasing automobiles vary by gender?

Answer:

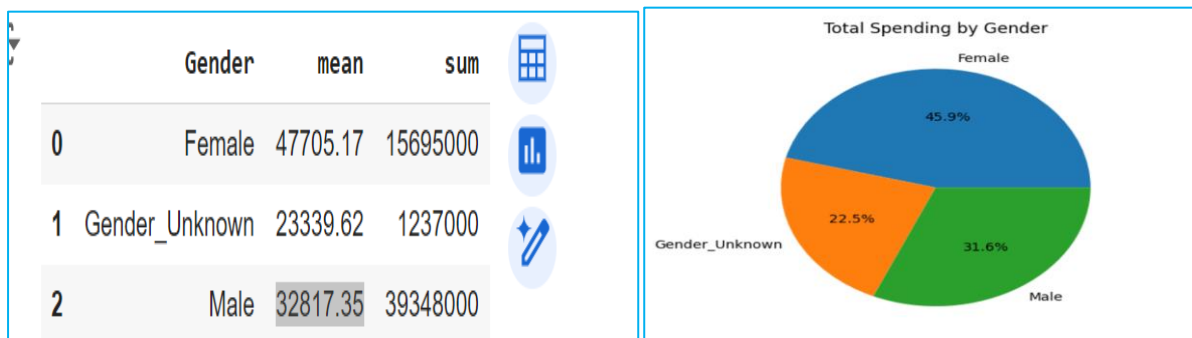


Figure 21 Total spending by Gender

Observation:

- On average, females spend more money on automobiles than males.
- Female spent more money on SUV as compare to male.

8.5 How much money was spent on purchasing automobiles by individuals who took a personal loan?

Answer:

Total amount was spent by customer who is having personal loan is 27,290,000.

8.6 How does having a working partner influence the purchase of higher-priced cars?

Answer:

	Partner_working	Price
0	No	47064.61
1	Yes	46335.70

Figure 22 Partner working vs Price

Observations:

- The customer who is spending the money having working partner is 46,335 (37.56%), whereas the customer who is spending the money not having working partner is 47,064 (38.29%).
- So, from above data we can conclude that having a working partner is not having much influence in the purchase of higher-priced cars.

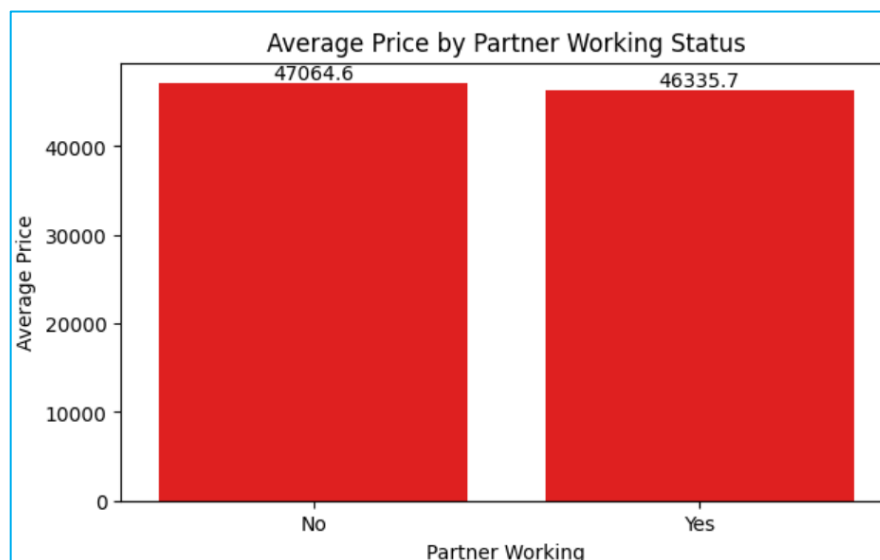


Figure 23 Average price by partner working status

Actionable Insights & Recommendations

9. Actionable Insights

Gender Preference:

- From above dataset, we can say that male customer shows very high preference for hatchback and Sedan. There is very least preference for SUV car.
- Whereas female customer shows very strong preference for the SUV Cars.

Age Preference:

- Customer in the age group 20 to 30 years shows more preference for hatchback and sedan and very less preference in SUV car.
- It is evident that customers in the age group 30-55 have gradually showing interest in SUV and there is drop in the interest towards hatchback.
- There is very less customer over 55 years of age, unable to catch the preference of cars

Profession:

- As per dataset the salaried customers are having very high preference for SUV car as compared to customers who are having business profession.
- Salaried and business professionals show equal interest in hatchback and for sedan.

Marital Status:

- Single customer shows more preference for hatchback
- Married customer shows mixed interest in all the car.

Total Salary:

- It is seen from the data set that SUVs are much more preferred by high income group of individuals while hatchback and sedans are preferred by wide range of income groups thus making those categories of cars more versatile when it comes to affordability.

Loan Impact:

- Customer having personal loans have a higher preference in sedan cars, while those without personal loans are more inclined towards Hatchbacks and SUVs.
- Individuals having a house loan have a more tendency to choose between Hatchback and SUVs.

Education:

- Customer who is having post graduate education are having more preference of SUV due to higher income as compare to customer is having Graduate education.

Number of Dependents:

- Customers with large family members have more tendency towards SUVs and sedan as these vehicles provide extra space and comfort, while smaller family size individuals prefer hatchback and sedan.

10. Recommendations

- **Age Advertisement:**

For young customers falling under the age group of 20-30 , we can create targeted campaign which focuses more on Hatchbacks and SUVs, while for individuals falling under the age group of 30-55, we need to target the campaign towards highlighting the features of SUV.

- **Gender Campaigns:**

For Male customers, we need to focus more towards highlighting the features of SUVs, while for female customers Hatchback and sedan should be focused.

- **Marital_Status Campaigns:**

For married customers we need to work towards family-oriented campaigns prioritizing Sedans and SUVs, and for unmarried customers we can go for lifestyle campaigns focussing Hatchbacks.

- **Salary Based Campaigns:**

For higher income customers we need to create campaign which focuses on luxury of SUVs, while emphasizing more on affordability points of Hatchback and sedans for broader range of income groups.

- **Loan_Status Based Campaigns:**

Customers who are having personal loans, then we should focus on the various financing options available and cost-effective cars like Sedans, while those without a loan we can focus on features of car and more on SUVs model, for customers having house loans, we can promote models like hatchbacks and sedans.

- **Education Based campaigns:**

Customers having education post graduate levels, we need to emphasize more on premium features of SUV, while for graduate levels we can emphasize on economic aspects of hatchbacks.

- **No_of _Dependents based Campaigns:**

If customers are having more no of dependents we can focus on safety and spacious features of SUV and sedan car.